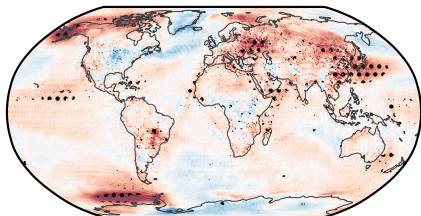


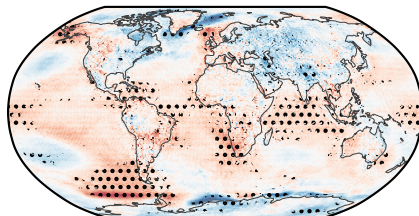
ENSEMBLE-MEAN emulator minus held-out CESM2 (5 emulator vs 6-11 CESM2 members), 10-year mean anomaly vs 1850-1900
hatching = difference significant at $p < 0.05$ (Welch t-test, FDR-controlled at $q = 0.1$); precipitation as % of its 1850-1900 baseline

Historical
2005-2014 (10 yr)



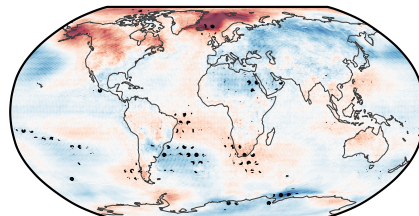
$r = 0.972$ RMSE = 0.183 °C
9% of area significantly different

SSP3-7.0
2091-2100 (10 yr)



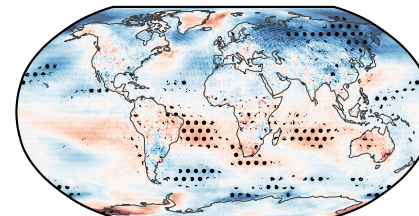
$r = 0.997$ RMSE = 0.164 °C
18% of area significantly different

Aerosol-only (AAER)
2041-2050 (10 yr)

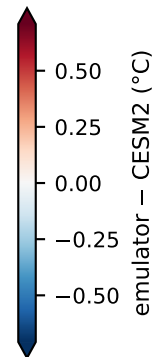


$r = 0.928$ RMSE = 0.160 °C
4% of area significantly different

Greenhouse-gas-only (GHG)
2041-2050 (10 yr)

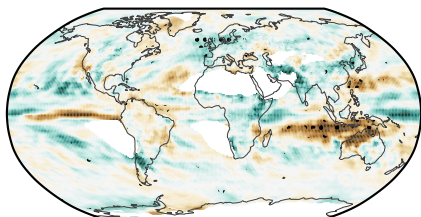


$r = 0.982$ RMSE = 0.214 °C
14% of area significantly different

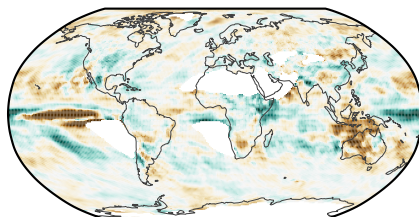


Temperature

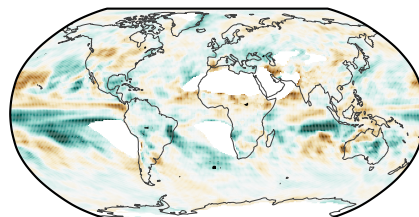
Precipitation



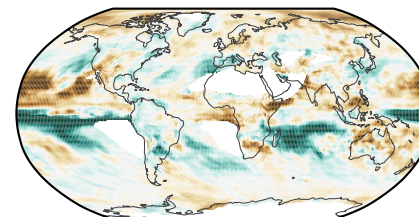
$r = 0.806$ RMSE = 4.953 %
2% of area significantly different
(9% arid, masked)



$r = 0.990$ RMSE = 5.322 %
0% of area significantly different
(9% arid, masked)



$r = 0.890$ RMSE = 4.493 %
0% of area significantly different
(9% arid, masked)



$r = 0.961$ RMSE = 7.016 %
0% of area significantly different
(9% arid, masked)

